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Rules

- The cover page should only have your name and email. Nothing else may be written on the cover page.
- No calculators, notes, books, or other outside resources with one exception: **While taking Exam 1, you may use the Standard Logical Equivalences reference sheet, which will be provided for you.**
- Eyes on your own exam.
- Scratch paper is available upon request. You may only use the provided scratch paper, and all scratch paper (used and unused) must be submitted with your exam.

Expectations

- Write with proper notation.
- Show your work in order to earn points. Show the process.

The exam length is 55 minutes. There are 100 possible points.

I wrote two sample answers, and how the rubric items [1], [2], [3], [4], and [5] apply.

(6pts) 1. Define even

Called $\heartsuit$ in rubric

Must match

Must match

The integer z is even if $\exists m \in \mathbb{Z}$ s.t. $z = 2m$.

[1] [2] [3] [4] [5] Called $\heartsuit$ in rubric

Must match

Must match

Let j be an integer. We say j is even if there exists an integer h such that $j = 2h$.

[1] [2] [3] [4] [5]

(6pts) 2. Define implication

The implication from proposition p to proposition q is the proposition $p \rightarrow q$ that is false when p is true and q is false, and true otherwise. We say "if p then q ."

(6pts) 3. Define converse

The converse of the implication $p \rightarrow q$ is the implication $q \rightarrow p$.

We can only take the converse of an implication.
(Can't do converse of a conjunction, or of a disjunction.)

Analogy: Reciprocal of a fraction? 😊 Reciprocal of a triangle? 😞

Sine of an angle? 😊 Sine of a Cartesian plane? 😞

Midpoint of a line segment? 😊 Midpoint of a function? 😞

(6pts) 4. Define proposition

A proposition is a declarative sentence that is true or false
but not both.

(6pts) 5. Define predicate

A predicate is a declarative statement which results in a proposition
after substituting x .

(6pts) 6. Define contradiction

A contradiction is a compound proposition that is always false,
regardless of the truth values of the propositions mentioned
therein.

Do not confuse contradiction with contrapositive.

Do not confuse contradiction with negation.

For all questions on this page:

- Let S be the set of all dragons.
- Let T be the set of all board games.
- Let $P(x, y)$ be the predicate x likes to play y

(4pts) 7. State what $m \in S$ means in plain English.

m is a dragon

(4pts) 8. State $\forall x \in S[\exists y \in T[P(x, y)]]$ in plain English.

All dragons like some board game.

(4pts) 9. State $\exists x \in S[\forall y \in T[P(x, y)]]$ in plain English.

There is a dragon that likes all board games.

(4pts) 10. State $\exists x \in S[\forall y \in T[\neg P(x, y)]]$ in plain English.

There is a dragon that doesn't like any board game.

(4pts) 11. State $\exists y \in T[\forall x \in S[P(x, y)]]$ in plain English.

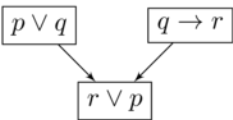
There is a board game that all dragons like.

(6pts) 12. If you know If Galinda golfs, then Boq bakes and you know Galinda golfs, what (if anything) can you conclude? Briefly explain.

Conclude: Boq bakes

Modus ponens

(6pts) 13. Show the following is a valid rule of inference:



p	q	r	$p \vee q$	$q \rightarrow r$	$r \vee p$
T	T	T	T	T	T
T	T	F	T	F	T
T	F	T	T	T	T
T	F	F	T	T	T
F	T	T	T	T	T
F	T	F	T	F	F
F	F	T	F	T	T
F	F	F	F	T	F

↑
only Ts

(6pts) 14. Show that $\boxed{q \rightarrow (p \vee q)}$ is a tautology without using a truth table.

$$\equiv \neg q \vee (p \vee q)$$

$$\equiv \neg q \vee (q \vee p)$$

$$\equiv (\neg q \vee q) \vee p$$

$$\equiv T \vee p$$

$$\equiv T$$

(6pts) 15. Negate and simplify until all negation symbol immediately precede propositions and predicates.

Show ALL steps.

$$\neg \forall a \in S [\exists b \in T [p \wedge Q(a, b)]]$$

$$\equiv \exists a \in S \neg \exists b \in T [p \wedge Q(a, b)]$$

$$\equiv \exists a \in S \forall b \in T \neg [p \wedge Q(a, b)]$$

$$\equiv \exists a \in S \forall b \in T [\neg p \vee \neg Q(a, b)]$$

(10pts) 16. Use the hypotheses:

- H1: If Obi opines, then Pam pays.
- H2: if Rex reads, then Dan dances.
- H3: If Pam pays, then Rex reads. ← *Never used.*
- H4: If Obi does not opine, then Leo leans.
- H5: if Dan dances, then Pam doesn't pay.

to prove: if Rex reads, then Leo leans.

The only ways to use H3

Pam pays

If Pam pays, then Rex reads

Rex doesn't read

Rex reads

Pam doesn't pay

But we never knew this.
In fact its negation is true.

Suppose Rex reads. WTS Leo leans. By H2 Dan dances. By H5, Pam doesn't pay.

By H1, Obi does not opine. By H4, Leo leans. $\square$

2nd answer

Alternate word: Suppose

Alternate word: Because
but not if

Alternate word could be knew
but not assume

Assume Rex reads. WTP Leo leans. Since Rex reads and H2 is true, we conclude Dan dances. Since H5 is true and Dan dances, we conclude Pam does not pay. Since Pam does not pay, we conclude Obi does not opine (by H1). We use the fact that Obi does not opine and use H4 together to conclude Leo leans. $\square$

alternate word: prove

(10pts) 17. Prove: if a is odd and b is odd, then $a - b$ is even.

Suppose a is odd and b is odd. So $\exists m \in \mathbb{Z}$ s.t. $a = 2m + 1$ and

$\exists n \in \mathbb{Z}$ s.t. $b = 2n + 1$. Let $z = m - n$. So $z \in \mathbb{Z}$ and

$$a - b = (2m + 1) - (2n + 1) = 2m + 1 - 2n - 1 = 2m - 2n = 2(m - n) = 2z.$$

So $a - b$ is even.

My work on line 217 proves that $a - b$ equals $2z$.

How? $a - b = (2m + 1) - (2n + 1)$ by substituting $a = 2m + 1$ and $b = 2n + 1$.

and $(2m + 1) - (2n + 1) = 2m + 1 - 2n - 1$ distributivity

and $2m + 1 - 2n - 1 = 2m - 2n$ simplifying $1 - 1$

and $2m - 2n = 2(m - n)$ factoring

and $2(m - n) = 2z$ substituting $m - n$ for z
because that is how I defined z here.